

| Component              | Component Pin   | Arduino Pin          | Notes / Rail                               |  |
|------------------------|-----------------|----------------------|--------------------------------------------|--|
| L298N Motor Driver     | IN1             | D4                   | Left Motor Direction 1                     |  |
|                        | IN2             | A0                   | Left Motor Direction 2                     |  |
|                        | IN3             | A1                   | Right Motor Direction 1                    |  |
|                        | IN4             | A2                   | Right Motor Direction 2                    |  |
|                        | ENA / ENB       | (Jumpers On)         | Standard full-speed enable jumpers kept on |  |
|                        | GND             | GND                  | Shared Common Ground                       |  |
|                        | 12V Terminal    | 12V Battery (+)      | Main Motor Power Rail                      |  |
| BMP180 Pressure Sensor | SDA             | A4                   | Hardware I2C SDA                           |  |
|                        | SCL             | A5                   | Hardware I2C SCL                           |  |
|                        | VCC             | 5V / 3.3V            | 3.3V or 5V (Check board regulator)         |  |
|                        | GND             | GND                  | Common Ground                              |  |
| Soil Moisture Sensor   | AO (Analog Out) | A6 (or A0 dependant) | Analog Soil Reading                        |  |
|                        | VCC             | 5V                   | 5V Power Rail                              |  |
|                        | GND             | GND                  | Common Ground                              |  |
| Rain Sensor            | AO (Analog Out) | A7                   | Analog Rain Intensity Reading              |  |
|                        | VCC             | 5V                   | 5V Power Rail                              |  |
|                        | GND             | GND                  | Common Ground                              |  |
| NEO-6M GPS Module      | TX              | D5 (SoftwareSerial)  | Transmits GPS data to Arduino              |  |
|                        | RX              | D6 (SoftwareSerial)  | Optional (Leave disconnected or to D6)     |  |
|                        | VCC             | 5V                   | 5V Power Rail                              |  |
|                        | GND             | GND                  | Common Ground                              |  |
| DHT11 Temp & Humidity  | DATA            | D2                   | Digital Sensor Signal                      |  |
|                        | VCC             | 5V                   | 5V Power Rail                              |  |
|                        | GND             | GND                  | Common Ground                              |  |
| Voltage Sensor         | S (Signal)      | A3                   | Battery Level Divider Input                |  |
|                        | GND / (-)       | GND                  | Common Ground                              |  |
| Pan Servo              | PWM Signal      | D9                   | Servo PWM Control                          |  |
|                        | VCC / GND       | 5V / GND             | Dedicated 5V Servo Rail                    |  |